Software Requirements Specification

For

University Course Management System, Release 1.0

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version
Hanya Essam Eldin	20/4/2026	Initial draft based on project description	1.0 Draft
Hanya Essam Eldin	27/4/2026	Updated requirements after elicitation process	1.1 Draft
Hanya Essam Eldin	8/5/2026	Final SRS document completed with diagrams and use case narratives	1.0 Final

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1.Introduction

1.1Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements for the University Course Management System (UCMS). It serves as a reference for all parties involved in the project, including the development team, testers, and project supervisors. The document defines what the system must do and how it must perform, forming the foundation for system design, implementation, and testing.

1.2 Project Scope

The University Course Management System (UCMS) is a web-based application built for a large academic institution. It brings all course-related activities together in one place, replacing the manual and disconnected processes currently used for course registration, assignment handling, and grade management.
The system serves four types of users: Administrator, Faculty Member, Registered Student, and Guest User. It is accessible from both desktop and mobile devices and is designed to be reliable, secure, and easy to use.
The system covers the following:

- Student course browsing and enrollment
- Assignment creation, submission, and grading
- Grade tracking and progress monitoring
- User and permission management
- Notification and reminder services

The system does not cover: financial transactions, live video streaming, physical classroom scheduling, or university admissions.

1.3 Definitions and Abbreviations

UCMS — University Course Management System.

SRS — Software Requirements Specification.

FR — Functional Requirement.

NFR — Non-Functional Requirement.

UC — Use Case.

GPA — Grade Point Average. A number that represents a student's overall academic performance.

PDF — Portable Document Format. A file format used for sharing documents.

Administrator — A system superuser who manages user accounts, roles, and course categories.

Faculty Member — An instructor or teaching assistant who creates courses, sets assignments, and tracks student progress.

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Registered Student — A student with an active account who can enroll in courses and submit assignments.

Guest User — A visitor who has not logged in. They can only view basic course information.

1.4 References

1. Project Description Document: "University Course Management System (UCMS)".
2. Requirements Elicitation Report

2. Overall Description

2.1 User Classes and Characteristics

The UCMS has four types of users. Each type has different permissions and uses the system in a different way.

Administrator The Administrator manages user accounts, roles, and course categories. They also make sure all data in the system is correct and consistent. This role requires a good understanding of system administration. Administrators do not use the system daily, but their actions have a big impact on the system.

Faculty Member The Faculty Member creates and manages courses, sets assignments, grades students, and monitors their progress. They are comfortable using digital platforms. Faculty members use the system regularly throughout the semester.

Registered Student The Registered Student browses and enrolls in courses, submits assignments, and views their grades. The system must be simple enough to use without any training. Students are the largest user group and use the system daily during the semester.

Guest User The Guest User can only view basic course information such as the course title, description, and category. They cannot enroll in courses or access any course content. No login is required. Guest users access the system occasionally .

2.2 Constraints, Assumptions, and Dependencies

Constraints

- The system must support at least 5,000 users at the same time without slowing down.
- All sensitive data such as passwords and grades must be encrypted.
- The system must work on both desktop and mobile browsers.
- Automated backups must run regularly to prevent data loss.
- Each user can only access the parts of the system that their role allows.

Assumptions

- All users already have a university-issued account before using the system.
- The university has a working email system that the UCMS can use.
- Users have access to a modern browser on a desktop or mobile device.
- Uploaded files will not exceed the system's maximum allowed size.

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Dependencies

- The UCMS depends on an email server to send confirmations, reminders, and grade notifications.
- The UCMS depends on a secure file storage service to save uploaded files and course materials.

3. Specific Requirements

3.1 External Interface

- The system shall work on common web browsers such as Chrome, Firefox, and Safari.
- The system shall work on both desktop computers and mobile devices.
- The system shall use an email service to send notifications and reminders to users.
- The system shall use a file storage service to save assignment files uploaded by students and faculty.
- The system shall be able to generate and download PDF reports for faculty members.
- The system shall use a secure connection to protect data during transfer.

3.2 Functional Requirements

3.2.1 Administrator:

- The system shall allow the administrator to manage user permissions. (Project description stated that Administrators are responsible for managing user permissions.)
- The system shall allow the administrator to create course categories.(Project description stated that Administrators are responsible for creating course categories.)
- The system shall allow the administrator to update course categories.(Project description stated that Administrators are responsible for updating course categories.)
- The system shall provide the administrator with tools to monitor, detect, and resolve data inconsistencies across all system records.(Project description stated that Administrators are responsible for ensuring data consistency and integrity across the system)

3.2.2 Faculty Member:

- The system shall allow the faculty member to create, edit and delete course content. (Project description stated that Faculty Members can create, edit, and delete course content.)
- The system shall allow the faculty member to create an assignment by either typing details directly on the platform or uploading a pre-prepared file, along with a deadline (Interview Question 1. Eng. Lamiaa Atef requested both options: typing or uploading. Prof. Nisreen Abotalib confirmed that file upload with a deadline is sufficient.)
- The system shall allow the faculty member to set multiple deadlines for a single assignment. (Interview Question 3. Both Eng. Lamiaa Atef and Prof. Nisreen Abotalib confirmed they need the ability to set multiple deadlines per assignment.)

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- The system shall allow the faculty member to export student progress reports as a PDF file. (Interview Question 4. Both Eng. Lamiaa Atef and Prof. Nisreen Abotalib confirmed they need to export student progress as a PDF report.)

- The system shall allow the faculty member to monitor student progress by viewing assignment submission grades, quizzes grades, and midterm exam of the current course and midterm exam ,final exam, GPA of the other courses. (Interview Question 5. Eng. Lamiaa Atef requested viewing grades within her own course only. Prof. Nisreen Abotalib additionally requested access to students' grades and GPA from other courses.)

- The system shall generate notifications for faculty member when assignment deadline has passed.(Interview Question 6. Both Eng. Lamiaa Atef and Prof. Nisreen Abotalib stated they want to be notified once the deadline has passed.)

- The system shall allow the faculty member to assign different assignments to different student groups within the same course.(Interview Question 2. Prof. Nisreen Abotalib stated she needs to assign different tasks to different student groups. Eng. Lamiaa Atef stated she does not need this feature, but it was included based on Prof. Nisreen's need.)

3.2.3 Registered Student:

- The system shall allow the registered student to browse the course catalog.(Project description stated that Registered Students can browse available courses)

- The system shall allow the Registered Student to enroll in an available course.(Project description stated that Registered Students can enroll available courses)

- The system shall allow the registered student to view instructor name and contact details, course schedule, course syllabus, list of assignments with deadlines, and instructor announcements on course page within the enrolled courses. (Questionnaire Question 10. 12 registered students via Google Forms were asked what information they need to see on a course page. Students listed all of these details [instructor contact, schedule, syllabus, assignments with deadlines, and announcements]as the most important things they need to find easily after enrolling in a course.)

- The system shall allow the registered student to submit assignments by uploading files before the specified deadline. (Project description stated that Registered Students can submit assignments.)

- The system shall allow the registered student to view their grades including individual assignment grades, quiz grades, midterm grade, final exam grade, final course grade, cumulative GPA, and GPA per semester also a faculty feedback for all graded assignments within their enrolled courses. (Questionnaire Question 9. 12 students via Google Forms listed all of these grade types as important to view in one place.)

- The system shall send an email confirmation to the registered student upon successful assignment submission. (Questionnaire Question 3. 12 students via Google Forms stated they need a confirmation to make sure their submission was received.)

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- The system shall send deadline reminder notifications to registered students at least 1 week and 24 hours before the assignment deadline. (Questionnaire Question 5. 12 students via Google Forms requested reminders at both 1 week and 24 hours before the deadline.)

- The system shall automatically save the student's assignment submission progress to prevent data loss in case of a page crash.(Questionnaire Question 7. 12 students via Google Forms reported losing work due to browser crashes and requested an auto-save feature.)

- The system shall record the assignment submission timestamp as the moment the student clicked Submit, regardless of server processing delay.(Questionnaire Question 6. 12 students via Google Forms requested this because they wanted the submission time to be recorded fairly, based on when they clicked Submit and not when the server processed it.)

- The system shall immediately notify the registered student when a faculty member posts their grade.(Questionnaire Question 8. 12 students via Google Forms stated they want to be notified immediately when a grade is posted.)

3.2.4 Guest User:

- The system allow the guest user to view general course information including course title, description, and category without requiring registration or login. (Project description stated that Guest Users are permitted to view general course information without enrolling.)

3.3 Non-Functional Requirements

3.2.1 Performance:

- The system must support 5000 concurrent users without performance degradation (Project description explicitly stated that the system must support up to 5,000 concurrent users without degradation in performance during peak times) High

- The system must respond to any user action within a maximum of 2 seconds (Interview Question 8. Eng. Lamiaa Atef stated that the system must respond within a maximum of 2 seconds. Prof. Nisreen Abotalib stated within 5 seconds. The stricter requirement of 2 seconds was adopted to satisfy both interviewees.) High

- The system shall confirm course enrollment within 5 seconds of the student clicking "Enroll."(Questionnaire Question 2. 12 registered students via Google Forms stated that confirmation for course enrollment within 5 seconds of the student clicking "Enroll.") High

- The system shall load the full grades page within 5 seconds.(Questionnaire Question 7. 12 registered students via Google Forms stated that full grade page must load within 5 seconds) Medium

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3.2.2 Security:

- All sensitive data should be encrypted (The project description stated that all sensitive data should be encrypted.) High

- Each user should only access the data permitted for them, 100% of unauthorized access attempts must be blocked(The project description stated that each user should only access the data permitted for them, and 100% of unauthorized access attempts must be blocked.) High

3.2.3 Reliability:

- The system shall perform automated data backups at least once every 24 hours to ensure data recovery in case of loss.(Project scope mentioned that regular backups are necessary to prevent data loss.) Medium

3.2.4 Usability:

- The course enrollment process must be completable without any guidance or training(Questionnaire Question 1. 12 registered students from via Google Forms stated that they need the course enrollment process to be easy) Medium

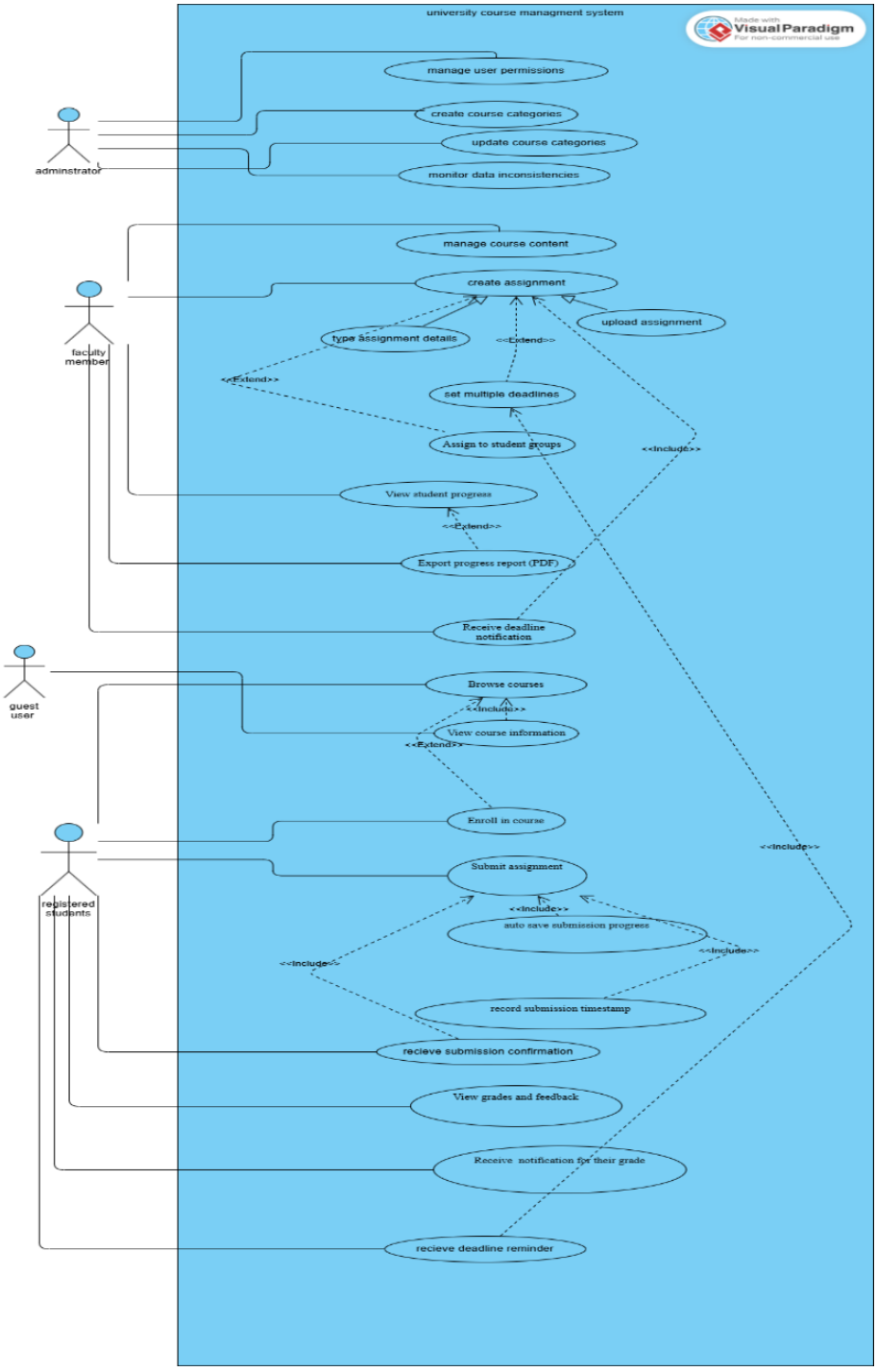
3.2.5 Portability:

- The system must be fully accessible on both desktop and mobile devices.(Interview Question 7. Eng. Lamiaa Atef stated she needs to access the system on both desktop and mobile. Prof. Nisreen Abotalib also stated she needs to access the system in both ways.) High

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4.Diagrams

4.1 Use case Diagram



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4.2 Use case Narratives

Use Case 1: Enroll in Course

<table>
<tr><td>Use Case Name:</td><td>Enroll in Course</td></tr>
<tr><td>Scenario:</td><td>A registered student successfully enrolls in an available course through the UCMS.</td></tr>
<tr><td>Triggering Event:</td><td>Student wants to register for a course and navigates to the Course Catalog.</td></tr>
<tr><td>Brief Description:</td><td>The registered student browses or searches for a course, verifies eligibility and is added to the course. The system confirms enrollment within 5 seconds and allows access to class list content.</td></tr>
<tr><td>Actors:</td><td>Registered Student</td></tr>
<tr><td>Related Use Cases:</td><td>None</td></tr>
<tr><td>Stakeholders:</td><td>Registered Student, Faculty Member, Administrator</td></tr>
<tr><td>Preconditions:</td><td>1. Student is logged into the UCMS.
2. At least one course is available in the catalog.
3. The enrollment period is currently open.</td></tr>
<tr><td>Postconditions:</td><td>1. Student is added to the class list.
2. The enrolled course appears on the student's dashboard.
3. Student can access all course content and assignments.</td></tr>
<tr><td>Flow of Activities</td><td>
<table>
<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. Student navigates to the Course Catalog.</td><td>1. System displays a list of all available courses (title, description, category).</td></tr>
<tr><td>2. Student selects a specific course.</td><td>2. System retrieves and displays the full course details.</td></tr>
<tr><td>3. Student clicks the Enroll button.</td><td>3.1 System validates whether the student meets enrollment criteria.
2. System adds the student to the class list.
3. System displays a confirmation message and updates the student dashboard within 5 seconds.</td></tr>
</table>
</td></tr>
<tr><td>Exception Conditions:</td><td>1. Course is full: System notifies student the course is at capacity. Enrollment is blocked.
2. Prerequisites not met: System displays the required courses the student has not completed.
3. Enrollment period closed: System notifies student that registration is not currently open.</td></tr>
</table>

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Use Case 2: Submit Assignment

Use Case Name:	Submit Assignment
Scenario:	A registered student uploads and submits a completed assignment file before the specified deadline.
Triggering Event:	Student opens an active assignment within a course and intends to submit their work.
Brief Description:	The student selects an assignment, uploads a file in an accepted format, and confirms submission. The system records the timestamp at the moment the student clicks Submit, auto-saves progress to prevent data loss, sends an email confirmation, and updates the assignment status to Submitted.
Actors:	Registered Student
Related Use Cases:	None
Stakeholders:	Registered Student, Faculty Member
Preconditions:	1. Student is logged into the UCMS. 2. Student is enrolled in the course containing the assignment. 3. The assignment deadline has not yet passed. 4. Assignment status is Not Submitted.
Postconditions:	1. Assignment status is updated to Submitted. 2. Submission timestamp is recorded as the moment the student clicked Submit. 3. An email confirmation is sent to the student. 4. Faculty member can view the submission in the student progress dashboard.
Flow of Activities	**Actor** / **System** 1. Student navigates to the course and selects the assignment. / 1.System displays assignment details: title, instructions, deadline, and accepted file formats. 2. Student uploads the assignment file. / 2.System automatically saves upload progress to prevent data loss in case of a page crash. 3. Student clicks the Submit button. / 3.1 System records the submission timestamp as the exact moment Submit was clicked. 2. System processes and stores the submitted file. 3. System updates the assignment status to Submitted. 4. System sends an email confirmation to the student. 5. System displays a success message on screen.
Exception Conditions:	1. Deadline passed: System blocks the upload and notifies the student that the deadline has passed. Submission is marked Late or rejected depending on faculty settings. 2.1 Invalid file format: System rejects the file and displays an error listing accepted formats. Student is prompted to re-upload. 2. Page crash during upload: System restores the auto-saved progress so the student can continue.

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Use Case 3: Create Assignment

Use Case Name:	Create Assignment
Scenario:	A faculty member creates a new assignment for students enrolled in one of their courses.
Triggering Event:	Faculty member wishes to add a new assignment to a course they are teaching.
Brief Description:	The faculty member selects a course and creates an assignment either by typing details directly on the platform or by uploading a pre-prepared file (PDF or Word document), along with one or more deadlines. The system optionally allows assigning the task to specific student groups. Upon saving, the system notifies enrolled students and schedules deadline reminder notifications.
Actors:	Faculty Member
Related Use Cases:	None
Stakeholders:	Faculty Member, Registered Student, Administrator
Preconditions:	1. Faculty member is logged into the UCMS. 2. Faculty member is the assigned instructor for the course. 3. The course has at least one enrolled student.
Postconditions:	1. Assignment is saved and published to the course. 2. Assignment is visible to all enrolled students (or to targeted student groups). 3. Enrolled students receive a notification about the new assignment. 4. Deadline reminder notifications are scheduled 1 week and 24 hours before the due date.
Flow of Activities	

Actor	**System**
1. Faculty selects a course from the dashboard and clicks Create Assignment.	1. System opens the assignment creation form.
2. Faculty enters the assignment title and instructions, OR uploads a pre-prepared PDF/Word file.	2. System accepts either typed content or the uploaded file and validates the file format.
3. Faculty sets one or more deadlines. Multiple deadlines may be defined for different student groups.	3. System validates that all deadlines are set to future dates.
4. (Optional) Faculty assigns the task to specific student groups within the course.	4. System applies the group restriction to the assignment.
5. Faculty clicks Save and Publish.	5.1 System saves the assignment and makes it visible to the targeted students. 2. System sends notifications to enrolled students about the new assignment.

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	3. System schedules deadline reminder notifications (1 week and 24 hours before).
Exception Conditions:	3. Invalid deadline: System alerts the faculty member that a deadline is set in the past. Faculty must correct the date before saving. 2. Invalid file format: System rejects the uploaded file and lists the accepted formats (PDF, Word). 5.1 No students enrolled: System displays a warning. Faculty may still save the assignment as a draft.

Use Case 4: View Student Progress

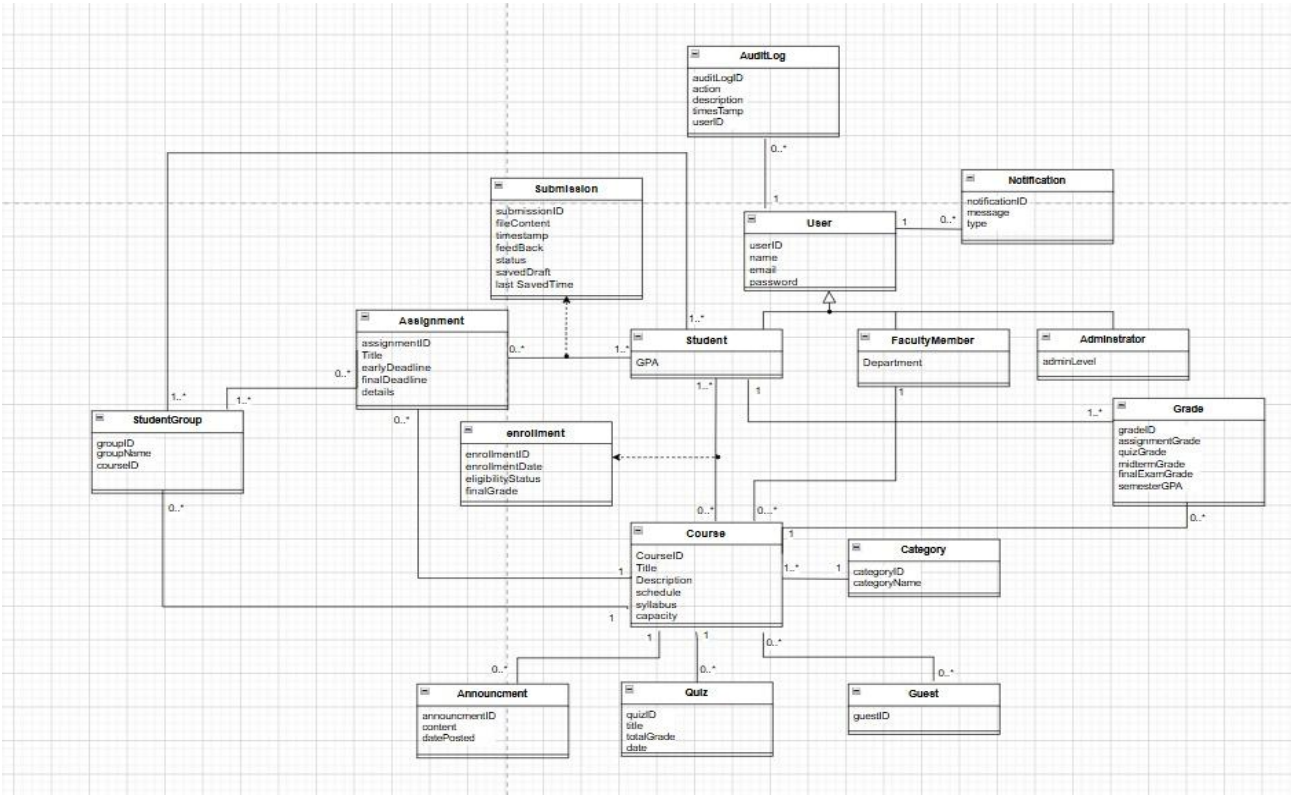
Use Case Name:	View Student Progress
Scenario:	A faculty member reviews the academic performance of students enrolled in one of their courses.
Triggering Event:	Faculty member wants to monitor student performance and clicks View Student Progress.
Brief Description:	The faculty member selects a course and accesses the student progress dashboard, which aggregates grades for assignments, quizzes, and midterm exams. Faculty with broader access (professors) may also view midterm, final exam grades, and GPA from other courses. The faculty member may optionally export the full progress report as a PDF file.
Actors:	Faculty Member
Related Use Cases:	None
Stakeholders:	Faculty Member, Registered Student, Administrator
Preconditions:	1. Faculty member is logged into the UCMS. 2. Faculty member is the assigned instructor for the selected course. 3. At least one student is enrolled with recorded activity in the course.
Postconditions:	1. Faculty member has reviewed up-to-date student progress data. 2. A PDF progress report has been generated and downloaded.(optional) 3. No data is modified so this is a read only operation.
Flow of Activities	

Actor	System
1. Faculty member selects a course from the dashboard.	1.1 System displays the course overview page.
2. Faculty clicks View Student Progress.	2.1 System aggregates student data: names, assignment submission grades, quiz grades, and midterm exam grades for the current course. 2. For faculty with broader access: System additionally displays midterm grades, final exam grades, and GPA from the student's other courses. 3. System displays the progress dashboard within 5 seconds.
3. (Optional) Faculty clicks Export Progress Report (PDF).	3.1 System generates a formatted PDF report containing all student progress data.

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		2. System initiates the file download for the faculty member.
Exception Conditions:	2.1No data available: System displays a placeholder message stating no student activity has been recorded yet for this course. 3.1 PDF export fails: System displays an error message and prompts the faculty member to retry. 2.2 System response exceeds 5 seconds: System shows a loading indicator and retries data retrieval.	

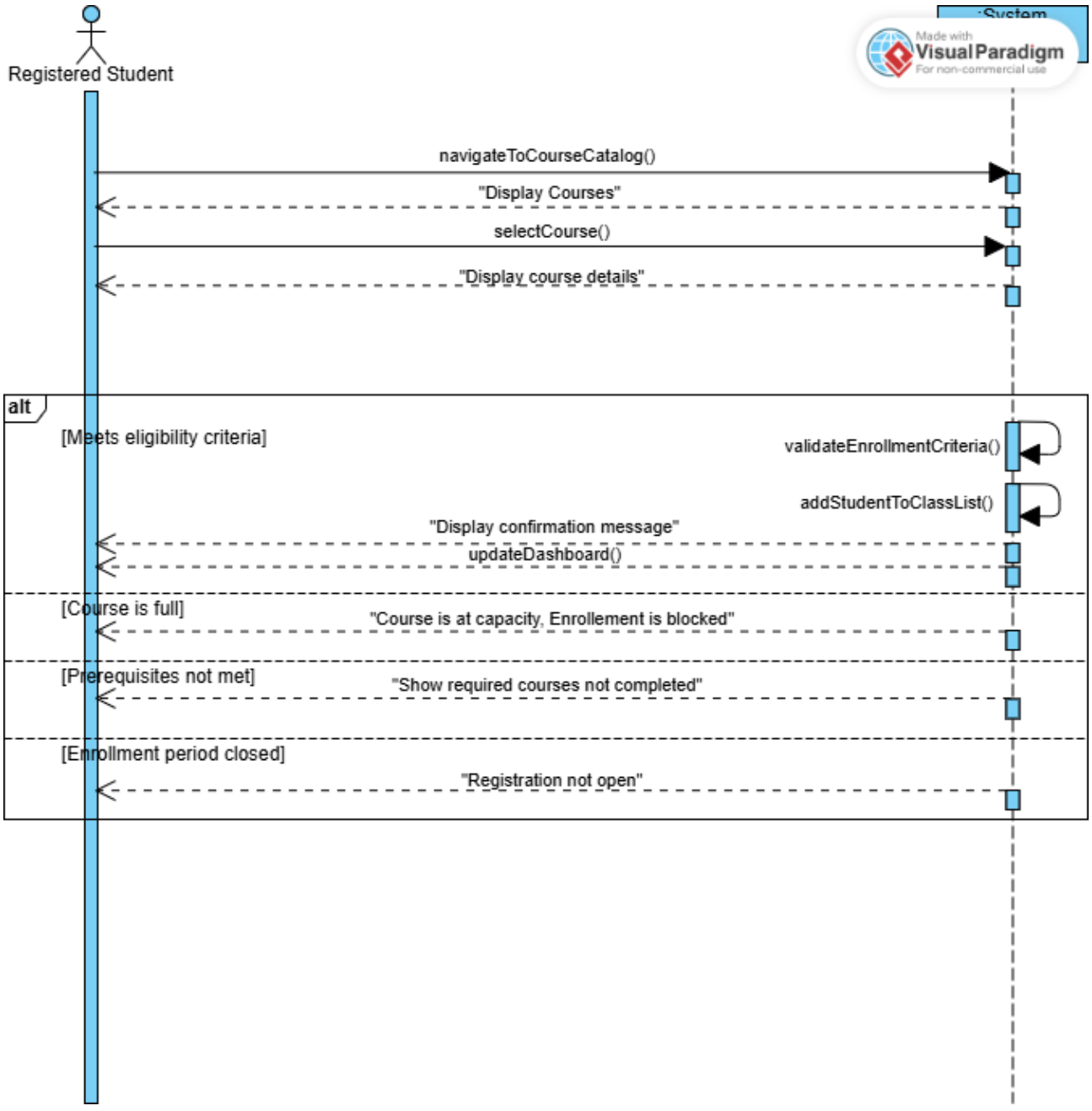
4.3 Class Diagram



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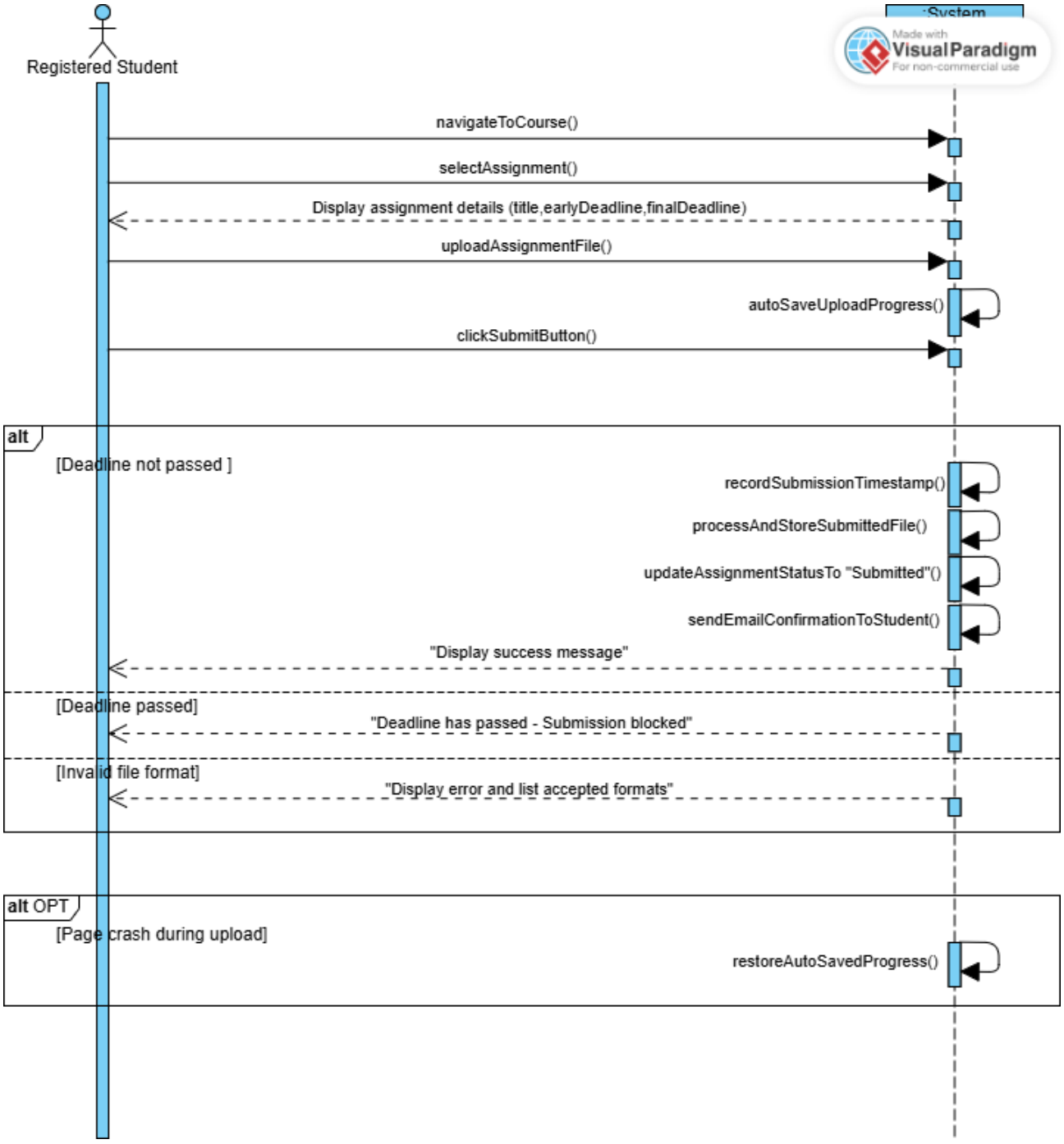
4.4 Sequence Diagrams

4.4.1 **Enroll in Course**



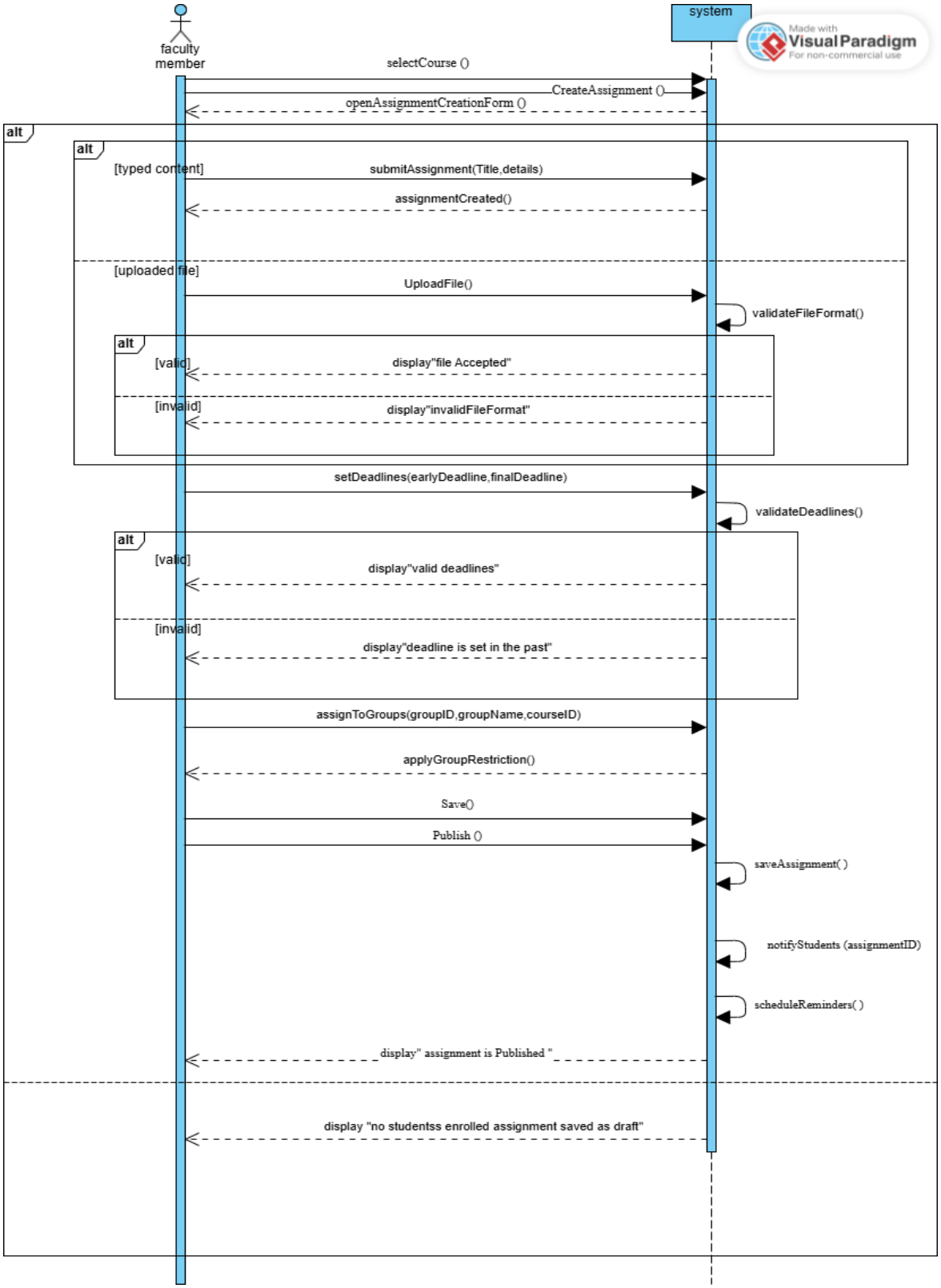
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4.4.2 **Submit Assignment**



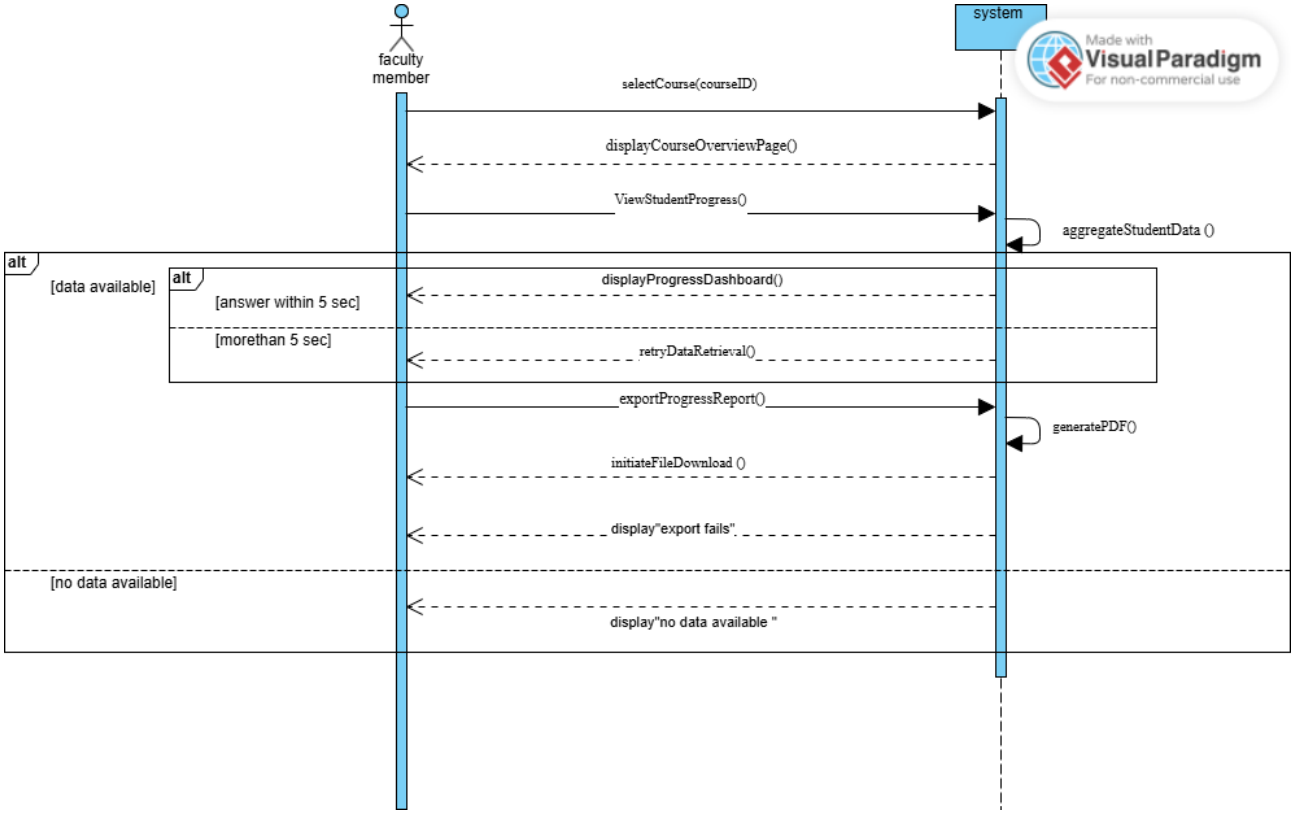
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4.4.3 **Create Assignment**



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4.4.4 **View Student Progress**



5. Additional Comments

5.1 Open Issues

ID	Issue Title	Date Identified	Target End Date	Responsible Project Person	User Contact	Comments
1	Maximum file size for uploads	April 27, 2026	May 12, 2026	Hanya Essam	Eng. Lamiaa Atef	What is the maximum file size allowed for assignment uploads?
2	Late submission policy	April 27, 2026	May 12, 2026	Hanya Essam	Prof. Nisreen Abotalib	Should late submissions be blocked, allowed, or flagged?

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5.2 Future Enhancements

- Integration with a live lecture or video-conferencing platform for real-time class sessions
- Online payment integration to allow students to pay course fees directly through the University Course Management System
.
- An AI-powered course recommendation engine that suggests relevant courses to students based on their academic history and interests.
- A physical classroom scheduling module that links with on-campus room booking systems.

5.3 Remaining Assumptions

- It is assumed that all users will interact with the system in English. Multilingual support is not addressed in this release.
- It is assumed that the questionnaire and interview findings represent the views of the broader target user population at the institution.
- The university already has the servers and technical infrastructure needed to run the system.

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